

Exercise 3 - Theoretical question #2: the resolution of the identity (density fitting)

The bottleneck

In HF and DFT the Coulomb and exchange matrices are built from four-centre two-electron integrals,

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma), \quad K_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\lambda|\nu\sigma), \quad (\mu\nu|\lambda\sigma) = \iint \phi_\mu(1) \phi_\nu(1) r_{12}^{-1} \phi_\lambda(2) \phi_\sigma(2) dr_1 dr_2.$$

Their number grows as N^4 with the number of basis functions N . For 2-chlorophenol with def2-TZVP ($N = 284$, Exercise 1) there are about 8.1×10^8 unique integrals, about 6.5 GB, and they have to be recomputed in every SCF iteration or read back from disk. Correlated methods (MP2, coupled cluster) must in addition transform them to the molecular-orbital basis, which costs N^5 .

The idea

A four-centre integral is the Coulomb interaction between two charge distributions, the orbital products $\rho_{\mu\nu}(r) = \phi_\mu(r) \phi_\nu(r)$ and $\rho_{\lambda\sigma}(r)$. The resolution of the identity (RI), also called density fitting, expands each product in an auxiliary basis of atom-centred functions $\{\chi_p\}$:

$$\rho_{\mu\nu}(r) \approx \sum_p c_{\mu\nu}^p \chi_p(r).$$

The coefficients are chosen to minimise the Coulomb self-energy of the fitting error, which gives $c_{\mu\nu} = V^{-1} (P|\mu\nu)$ with the Coulomb metric $V_{pQ} = (P|Q)$. With this choice the error in the energy is only second order in the error of the fitted density. The four-index integral then factorises into three- and two-index pieces:

$$(\mu\nu|\lambda\sigma) \approx \sum_{pQ} (\mu\nu|P) [V^{-1}]_{PQ} (Q|\lambda\sigma).$$

The name comes from this step: it is the same as inserting an approximate identity operator, $1 \approx \sum_{pQ} |P\rangle [V^{-1}]_{PQ} \langle Q|$, between the two charge distributions. The quality is controlled by the auxiliary basis; ORCA provides fitted sets for each purpose, e.g. def2/J for Coulomb, def2/JK for Coulomb and exchange, def2-TZVP/C for correlation, SARC/J for relativistic basis sets, or AutoAux to generate one automatically.

Where the speed-up comes from

Far fewer integrals. Only three-centre $(\mu\nu|P)$ and two-centre $(P|Q)$ integrals are needed, and the auxiliary basis is only 1.5–3 times larger than the orbital basis (def2/J: $N_{\text{aux}} = 449$ for 2-chlorophenol). The count drops from about $N^4/8 = 8.1 \times 10^8$ to $N^2 N_{\text{aux}}/2 = 1.8 \times 10^7$, roughly 45 times fewer, and at about 145 MB they can be kept in memory instead of being recomputed or read from disk.

Coulomb matrix in two cheap steps (RI-J). First the density is fitted, $d = V^{-1} g$ with $g_Q = \sum_{\lambda\sigma} (Q|\lambda\sigma) P_{\lambda\sigma}$; then $J_{\mu\nu} = \sum_p (\mu\nu|P) d_p$. Both steps cost $N^2 N_{\text{aux}}$ instead of N^4 and are dense matrix–vector operations that run efficiently on modern processors. For GGA functionals such as PBE, J is the only two-electron term, so RI-J removes the main bottleneck of the whole calculation. The fitting error is small and systematic, so it largely cancels in energy differences.

Exchange (RIJK and RIJCOSX). The exchange matrix couples the indices crosswise and does not factorise as favourably. ORCA either fits it as well (RIJK, with a /JK auxiliary basis) or uses RIJCOSX: RI for J and the chain-of-spheres method for K, in which one electron coordinate is integrated analytically and the other on a numerical grid, $K_{\mu\nu} \approx \sum_g w_g \phi_\mu(r_g) \sum_{\lambda\sigma} A_{\nu\lambda}(r_g) P_{\lambda\sigma} \phi_\sigma(r_g)$, $A_{\nu\lambda}(r_g) = \int \phi_\nu(r) \phi_\lambda(r) |r - r_g|^{-1} dr$. This scales close to linearly for large molecules. In Exercise 1 the PBE runs used RI-J (def2/J) and the B3LYP runs RIJCOSX (def2/J).

Correlated methods. In RI-MP2 the MO integrals are factorised in the same way, $(ia|jb) \approx \sum_p B_{ia}^p B_{jb}^p$, which removes the expensive four-index transformation and gives speed-ups of an order of magnitude or more. The same factorisation is used throughout ORCA's DLPNO methods.